CSS : Cascading Style Sheets

To add visual appeal and enhance the user experience of a website.

```html
<!DOCTYPE html>
<html>

<head>
  <title>My Website</title>
</head>

<body>
  <p style="color:blue">Greetings
and welcome to the CSS course. I am
a p tag that has been styled using
CSS to appear in a pleasing shade of
blue.</p>
</body>

</html>
```

Greetings and welcome to the CSS course. I am a p tag that has been styled using CSS to appear in a pleasing shade of blue.

In Html , we can create only Black and White Webpages

So ,In CSS we will make webpages visually appealing by customizing

1)CSS Syntax:

Selector{CSS property name: Declaration 1;
CSS property name: Declaration 2;
}

Selectors: p,h1,id,class

Declaration: contains one or more properties . Property name and Property valueare the two sections that make up a declaration part in CSS

This is done between **<style> </style>**

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
  <style>
    p {
      color: blue;
      font-size: 28px;
    }
  </style>
</head>

<body>
  <p>I am a paragraph that will be
displayed in blue with a font size of
28px using CSS.</p>
</body>

</html>
```

I am a paragraph that will be displayed in blue with a font size of 28px using CSS.

2)ID Attributes: Unique Identifier
We cannot create 2 same para in html
In css, we use #ID

The HTML ID attribute is used to specify a **unique identifier** for an HTML element.
The ID must be **unique** with in the HTML document.
In CSS, the **#symbol** is used to select an element by its ID.

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
  <style>
    #main-para {
      color: blue;
    }
  </style>
</head>

<body>
  <p id="main-para">I am a paragraph
with the ID attribute set to 'main-
para', and I will be displayed in blue
using the CSS ID selector.</p>
</body>

</html>
```

I am a paragraph with the ID attribute set to 'main-para', and I will be displayed in blue using the CSS ID selector.

3)Class Attributes: Group HTML elements with same name share common characteristics.
Dot symbol is used
Mutliple HTML elements can have same class name.

```html
<!DOCTYPE html>
<html>
<head>
  <title>My Website</title>
  <style>
    .notes {
      font-style: italic;
      font-weight: bold;
    }
  </style>
</head>

<body>
  <h1>HTML Basics</h1>

  <p class="notes">Note: The head
element of an HTML document is the
part that is not displayed in the web
browser when the page is loaded.</p>

  <p>The body element contains all the
contents of an HTML document, such as
headings, paragraphs, images,
hyperlinks, tables, lists, etc.</p>
</p>
</body>
</html>
```

HTML Basics

Note: The head element of an HTML document is the part that is not displayed in the web browser when the page is loaded.

The body element contains all the contents of an HTML document, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.

4)CSS Selectors:

Universal selector (*) selects all html elements and apply style to each one
Element selector(p,h1..): Based on the element name.
ID selector(#): Based On its ID
Class selector(.): Based on its Class
We can group selectors if they have **same style**

```html
<!DOCTYPE html>
<html>

<head>
  <title>My Website</title>
  <style>
    * {
      color: green;
    }

    p {
      color: violet;
    }

    #id-selector {
      color: orange;
    }

    .class-selector {
      color: blue;
    }

    h3,
    h4 {
      color: red;
    }
  </style>
</head>
```

CSS Selectors

The universal selector (*) selects all HTML elements on the page.

Element Selector

The element selector selects HTML elements based on their element name.

Id Selector

In CSS, we use the # symbol to select the element based on its ID.

Class Selector

In CSS, we use the '.' symbol to select the

```html
<body>
  <h1>CSS Selectors</h1>
  <h2> The universal selector (*) selects all HTML elements on the page. </h2>
  <hr>
  <h2>Element Selector</h2>
  <p>The element selector selects HTML elements based on their element name.</p>
  <hr>
  <h2>Id Selector</h2>
  <p id="id-selector">In CSS, we use the # symbol to select the element based on its ID.</p>
  <hr>
  <h2>Class Selector</h2>
  <p class="class-selector">In CSS, we use the '.' symbol to select the element based on its class.</p>
  <p class="class-selector">In CSS, we use the '.' symbol to select the element based on its class.</p>
  <hr>
  <h2>Grouping Selector</h2>
  <h3>Heading 3</h3>
  <h4>Heading 4</h4>
</body>
```

The element selector selects HTML elements based on their element name.

Id Selector

In CSS, we use the # symbol to select the element based on its ID.

Class Selector

In CSS, we use the '.' symbol to select the element based on its class.

In CSS, we use the '.' symbol to select the element based on its class.

Grouping Selector

Heading 3

Heading 4

</html>

5)Cascading Order:

The styles cascade into new styles based on their priority.multiple styles are defined for the same HTML element in a web page

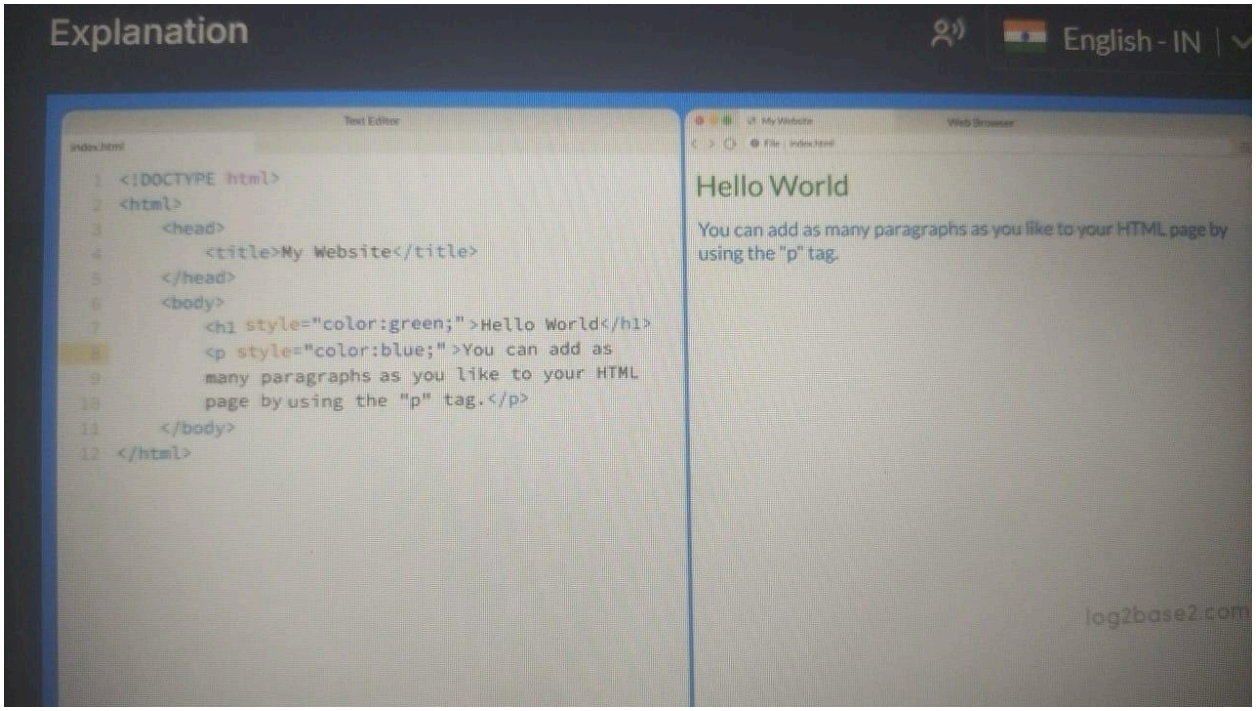
Inline > internal > external > browser default

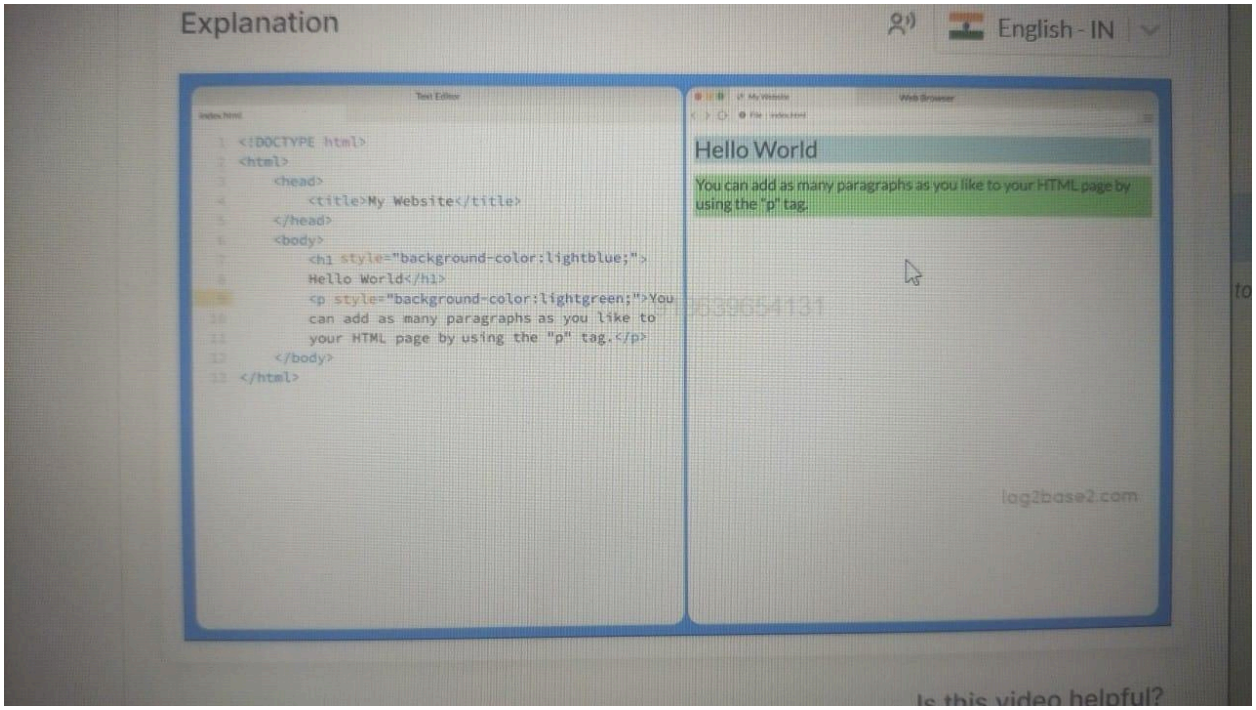
6)How to insert CSS

1. **Inline :**Unique style for One HTML element
2. **Internal:**if multiple pages are there and we need Unique html styles for single element
   Defined in style within Head section (Unique style for Single HTML page)
3. **External :** if we save this file with .css extension and included with a link element in Head section . (common style for all the pages)
   If we make any changes in style , it will be reflected to all HTML pages.

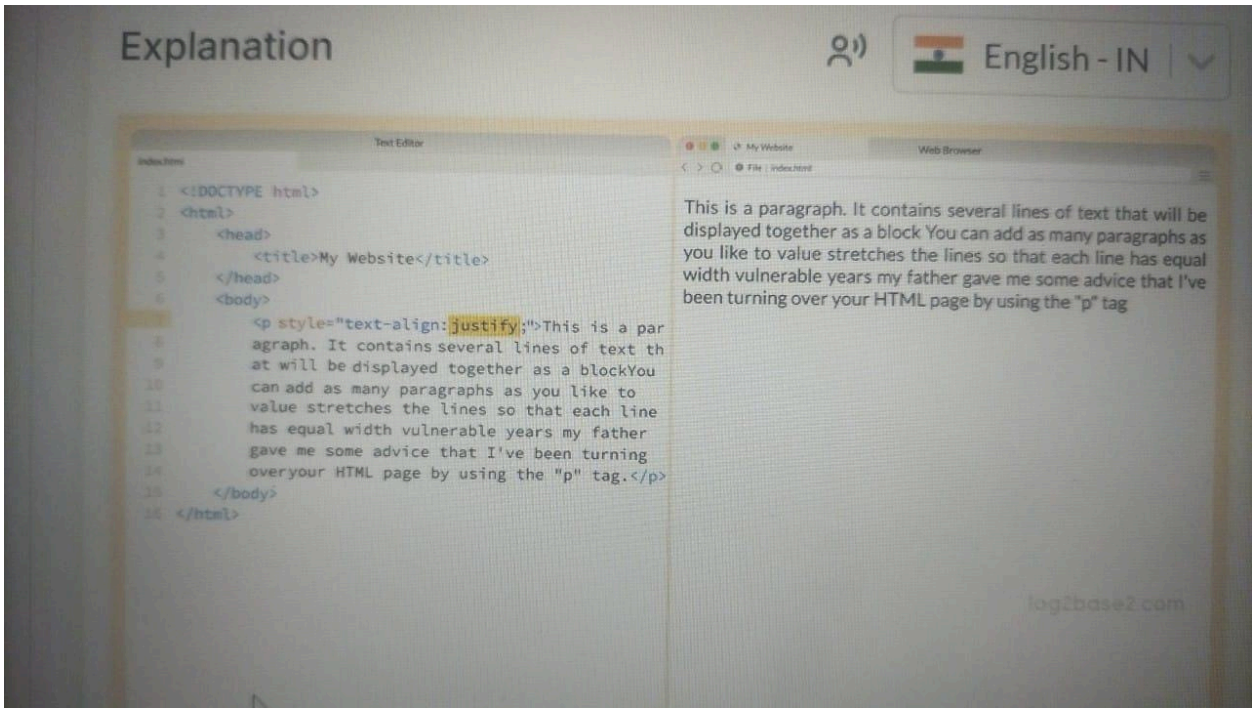
rel = relation between link resource and current Document

7)Text Color & Back ground color

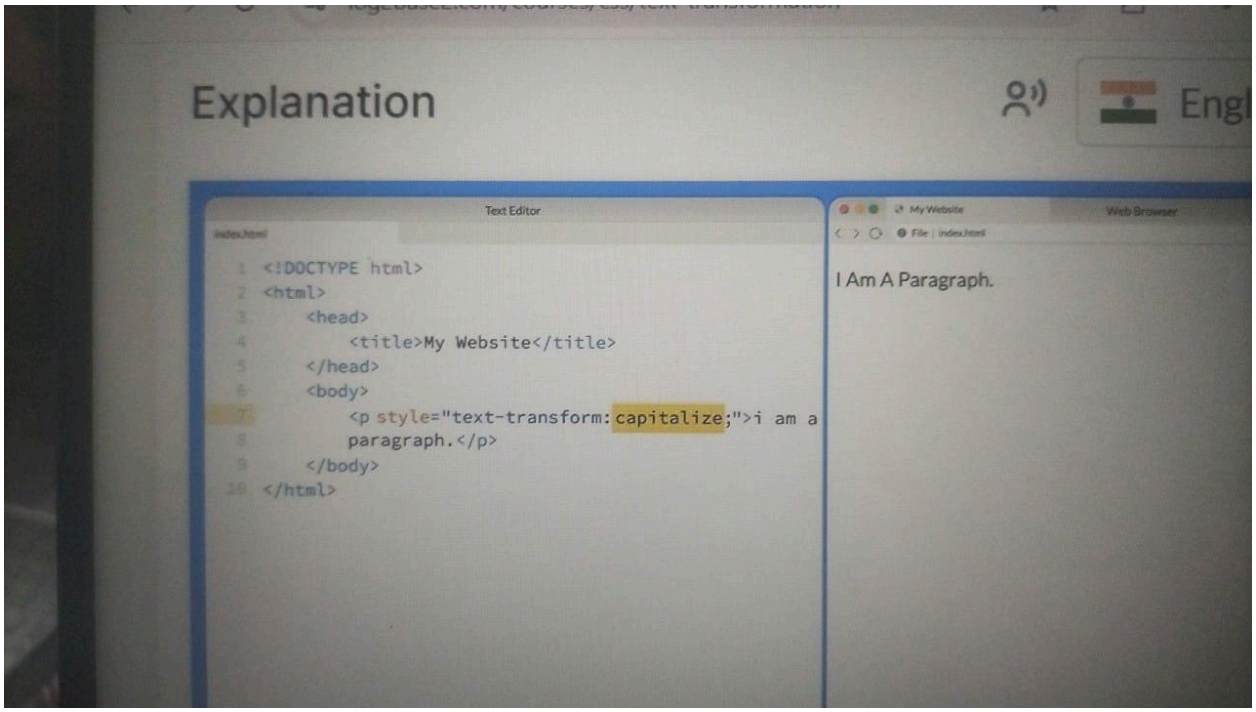




8)Text Alignment: right, left , center ,justify : to remove spaces

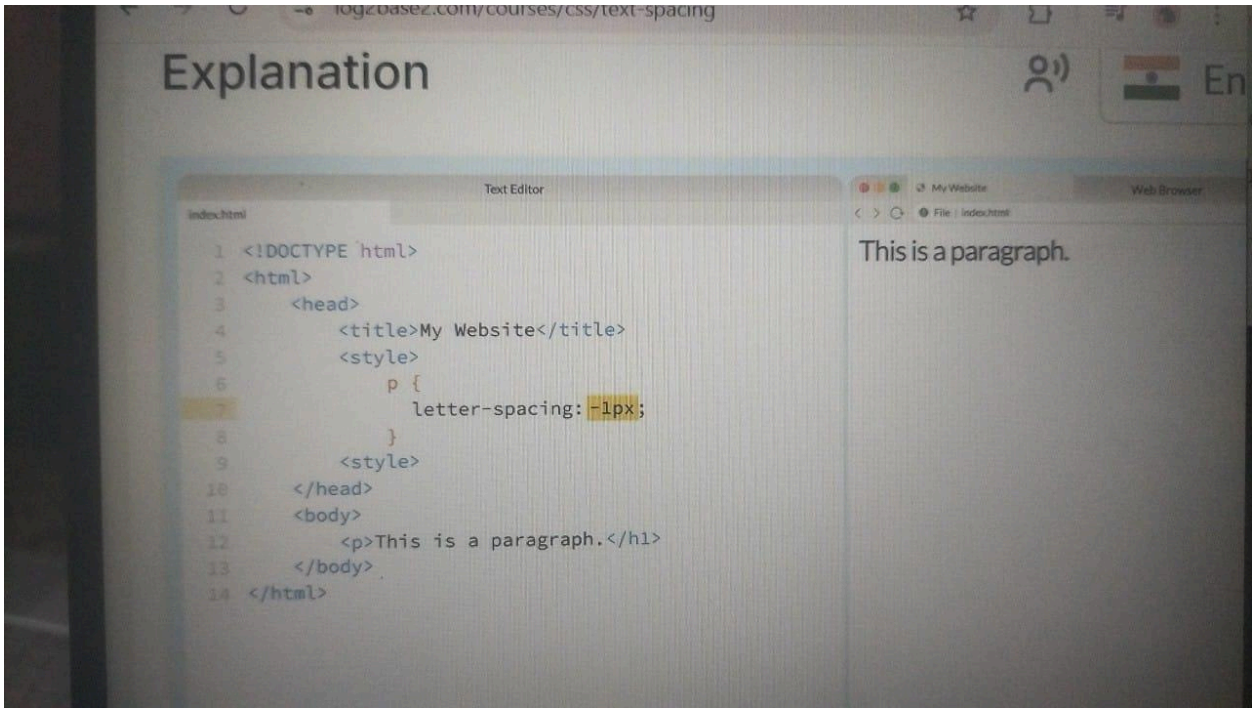


9) Text Transformation: transform in to uppercase or lower case letters

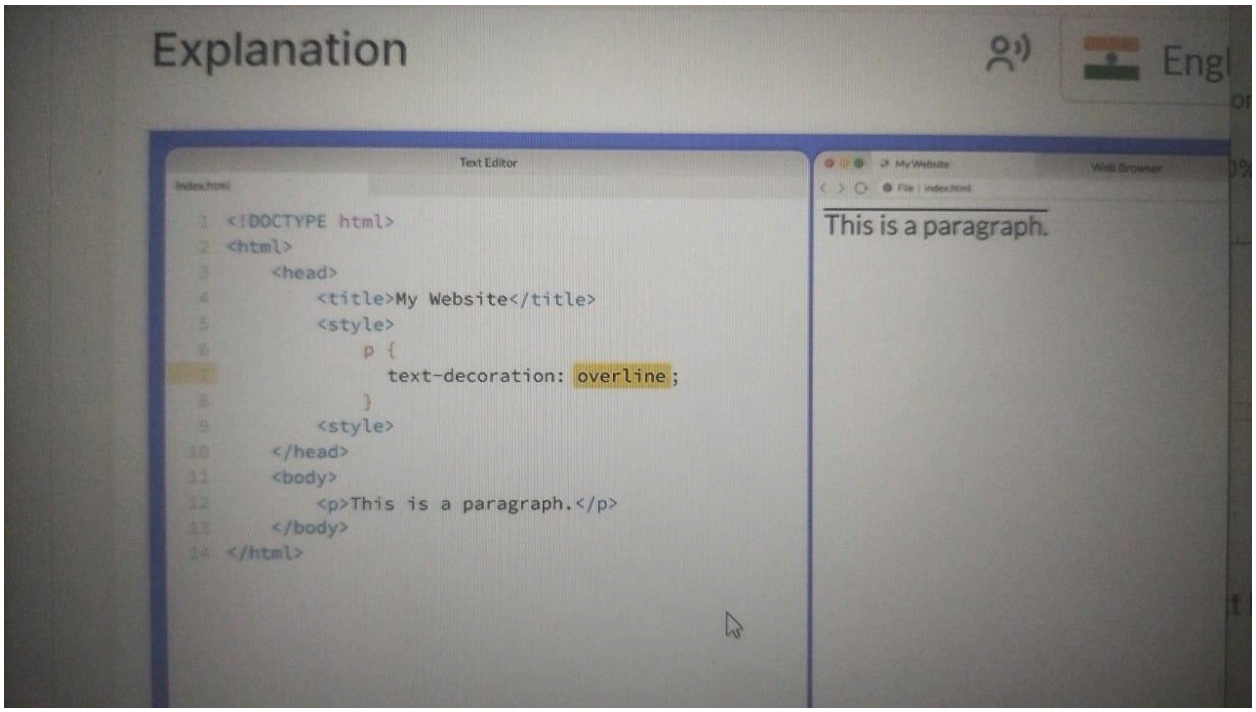


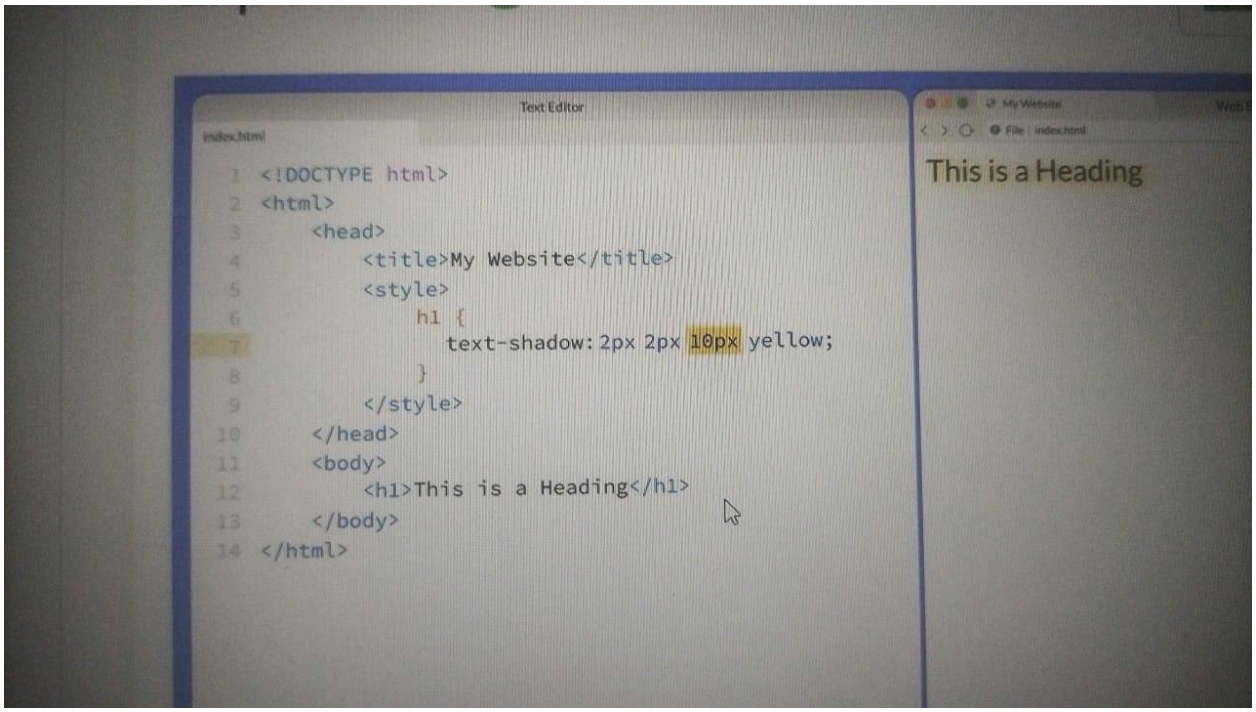
10)Text Spacing: Text Indentation , Word spacing ,Letter spacing , Line Height

Letter spacing: 2px,-1px
Word spacing:5 px,
Line height: > 1.1-1.2,<
Text Indentation :25px (first line will move a bit)



11) Text Decoration and Shadow:underline , overline,line-through





12)Font: Times new roman ,arial so on

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <p>This paragraph is currently
displayed in browser default font.</p>
  <p style="font-family:Arial;">This
paragraph is currently displayed using
the Arial font.</p>
  <p style="font-
family:Helvetica;">This paragraph is
currently displayed using the Helvetica
font.</p>
  <p style="font-family:'Times New
Roman';">This paragraph is currently
displayed using the Times New Roman
font.
  </p>
  <p style="font-family:'Courier
New';">This paragraph is currently
displayed using the Courier New font.
</p>
</body>
```

This paragraph is currently displayed in browser default font.

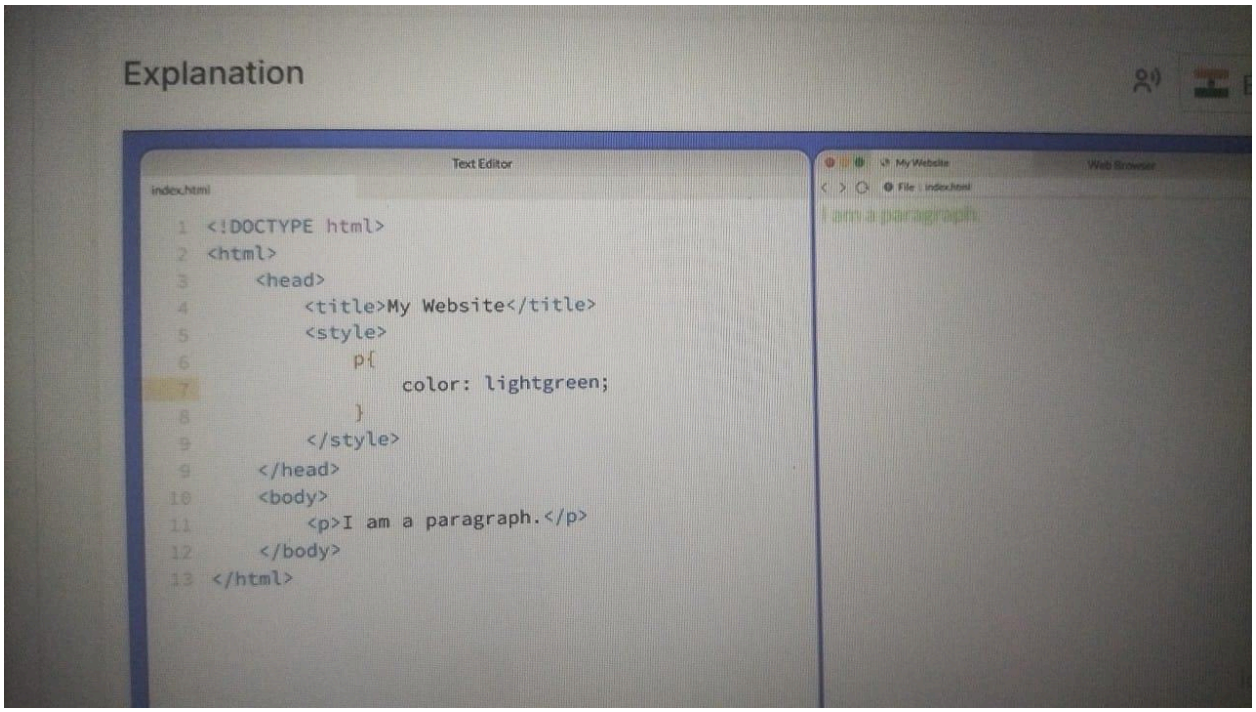
This paragraph is currently displayed using the Arial font.

This paragraph is currently displayed using the Helvetica font.

This paragraph is currently displayed using the Times New Roman font.

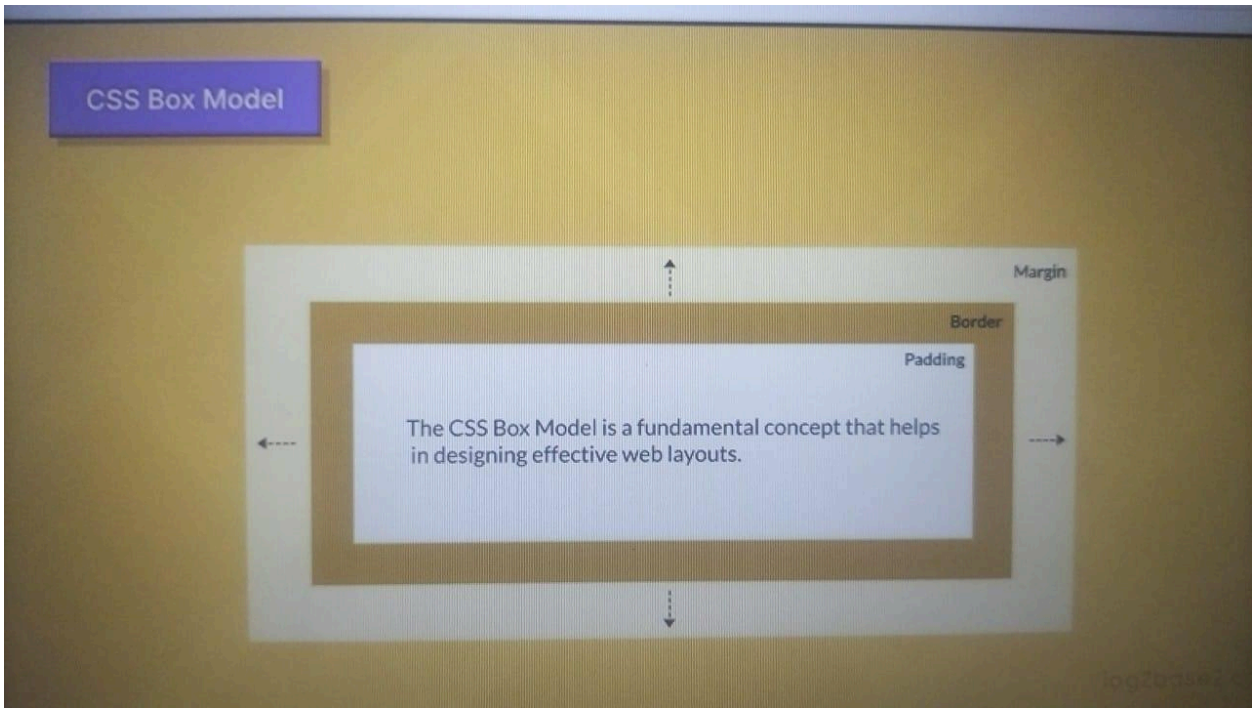
This paragraph is currently displayed using the Courier New font.

13)CSS Colors:



RGB

14)Box Model:

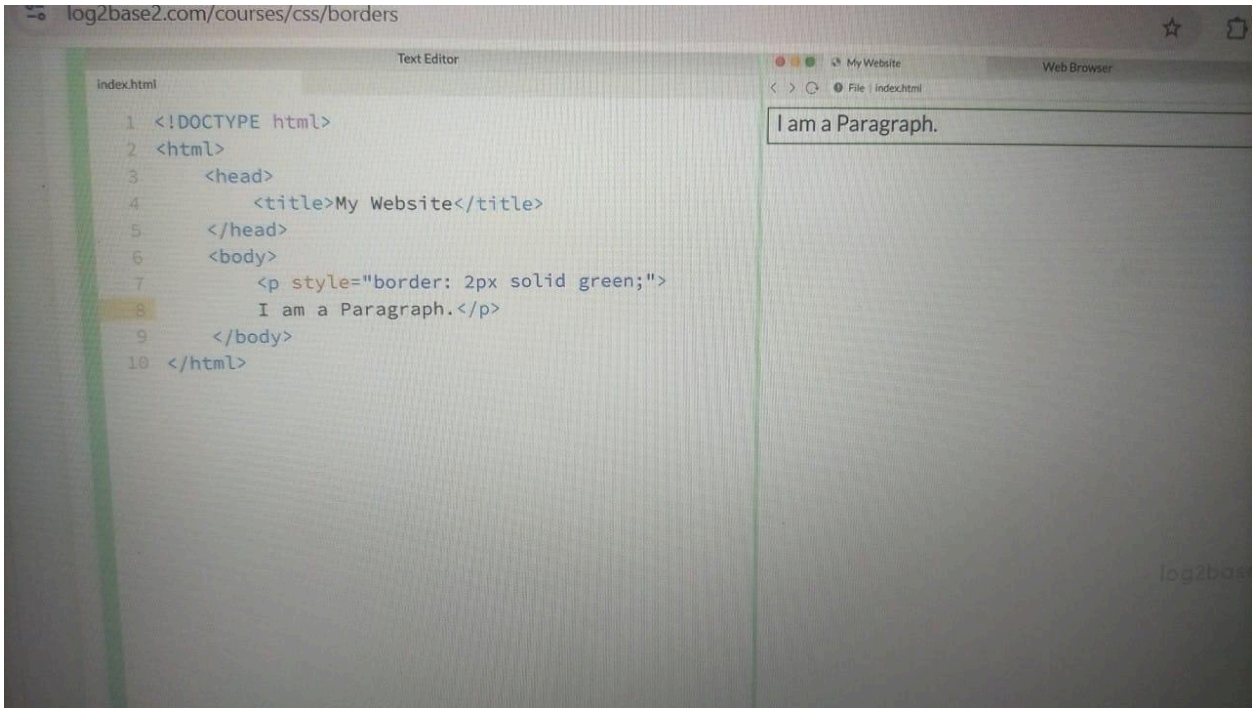


LIVE

```
<!DOCTYPE html>
<html>

<head>
  <title> CSS Box Model </title>
  <style>
    p {
      width: 250px;
      background-color: lightblue;
      ;
      padding: 20px;
      border: 8px solid blue;
      margin: 10px;
    }
  </style>
</head>

<body>
  <p> The CSS Box Model is a fundamental
concept that helps in designing
effective web layouts. </p>
</body>

</html>
```

The CSS Box Model is a fundamental concept that helps in designing effective web layouts.

15) Border width:



16)Background:

```
<!DOCTYPE html>
<html>

<head>
  <title> My Website </title>
  <style>
    .ex1 {
      background-color: lightblue;
      background-image:
url("https://log2base2.com/Assets/tree.
png");
      background-repeat: no-repeat;
      background-attachment: fixed;
      background-position: top right;
    }

    .ex2 {
      background: lightblue
url("https://log2base2.com/Assets/tree.
png") no-repeat fixed left;
    }
  </style>
</head>

<body>
  <div class="ex1">
    <h1> CSS Background </h1>
    <p> Lorem ipsum dolor sit amet,
```

</p> </body></html>

Repeat-x : horizontal , y:vertical , no : one ,attached : image fixed; text scrolled

17)Padding:

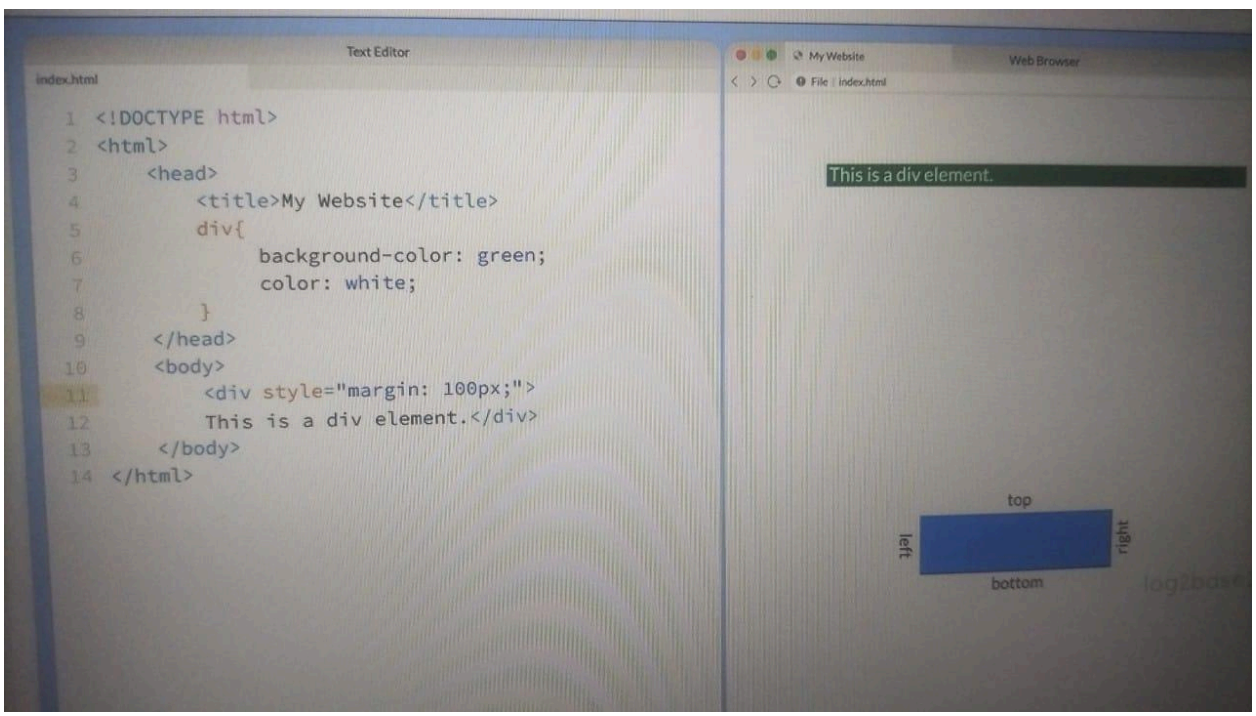
```html
<!DOCTYPE html>
<html>

<head>
  <style>
    div {
      background-color: blue;
      color: white;
    }
  </style>
</head>
<body>
  <div style="padding-top:10px;
            padding-right:10px;
            padding-bottom:10px;
            padding-left:10px;">This div
element has 10px padding on all sides
using the padding-* property.</div>
  <br>
  <div style="padding:10px;">This div
element has 10px padding on all sides
using shorthand padding property.</div>
</body>

</html>
```

LIVE

This div element has 10px padding on all sides using the padding-* property.

This div element has 10px padding on all sides using shorthand padding property.

18) Margin:

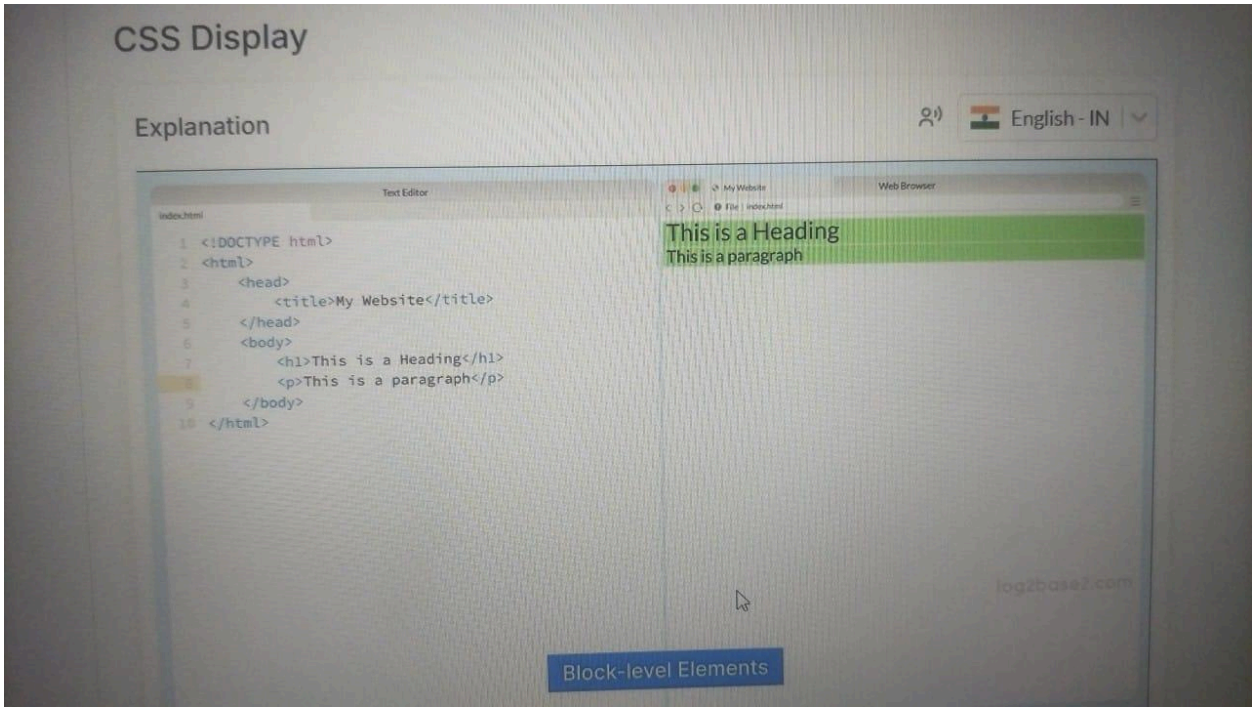


19)Height & width:

```
<!DOCTYPE html>
<html>

<head>
  <title> My Website </title>
  <style>
    div {
      background-color: lightgreen;
      width: 500px;
      height: 200px;
    }
  </style>
</head>

<body>
  <div> This is sample div element.
</div>
</body>

</html>
```

This is sample div element.

20) CSS Display: block elements , inline elements, overriding the default display values



Text Editor

index.html

```html
<!DOCTYPE html>
<html>
    <head>
        <title>My Website</title>
        <style>
            a{
                display: none;
            }
        </style>
    </head>
    <body>
        <a href="https://www.google.com">Google
        </a>
        <a href="https://www.wikipedi.org">Wikipedia
        </a>
    </body>
</html>
```

My Website

Web Browser

File | index.html

Google Wikipedia

Explanation

English

Text Editor

index.html

```html
<!DOCTYPE html>
<html>
    <head>
        <title>My Website</title>
        <style>
            a{
                display: block;
            }
        </style>
    </head>
    <body>
        <a href="https://www.google.com">Google
        </a>
        <a href="https://www.wikipedi.org">Wikipedia
        </a>
    </body>
</html>
```

My Website

Web Browser

File | index.html

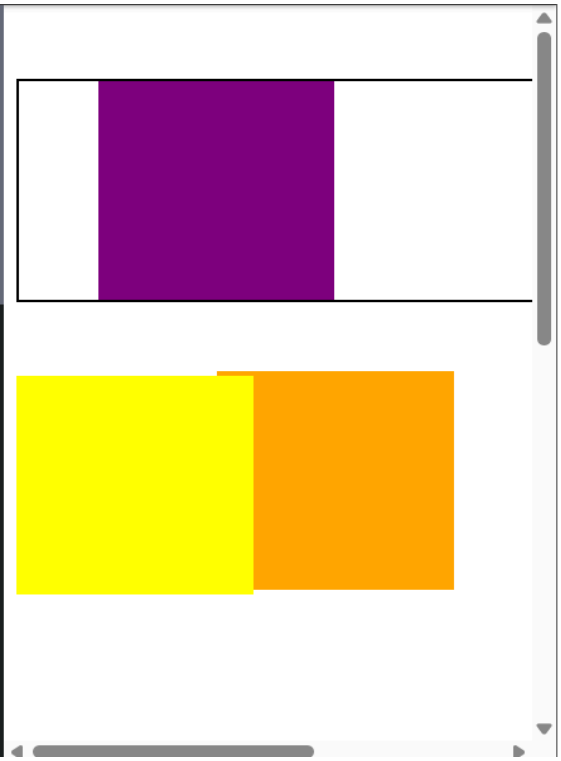
Google

Wikipedia

21)CSS Position:static,absolute, relative ,sticky,fixed

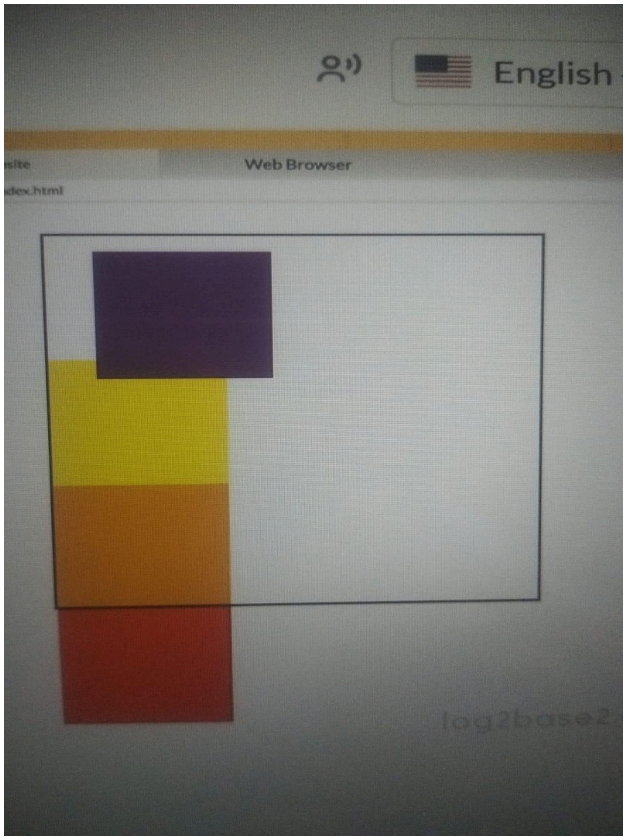
```html
<!DOCTYPE html>
<html lang="en">

<head>

  <style>
    body {
      height: 200vh;
    }

    .container {
      width: 550px;
      border: 2px solid black;
      margin: 50px auto;
      position: relative;
    }

    .box {
      width: 150px;
      height: 150px;
    }

    /*  Relative  */
    .purple {
      background-color: purple;
      position: relative;
      left: 50px;
    }
```

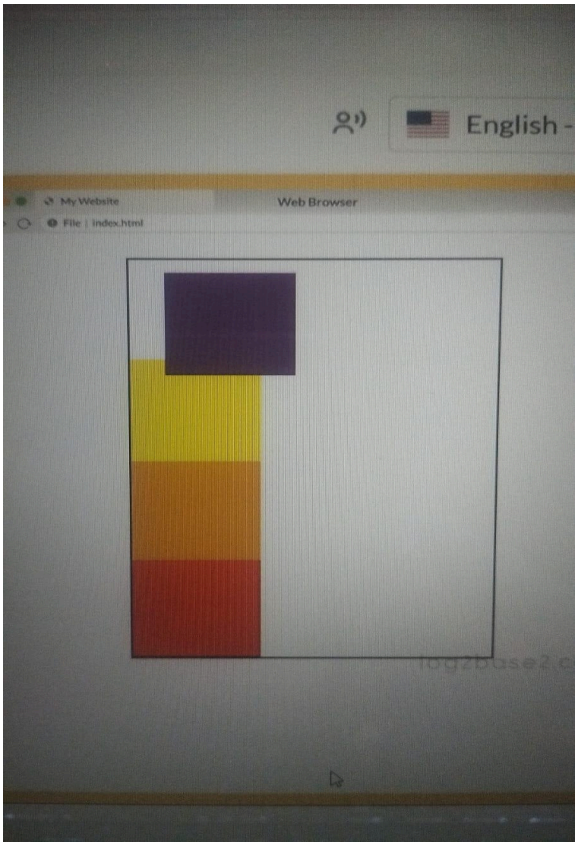
LIVE



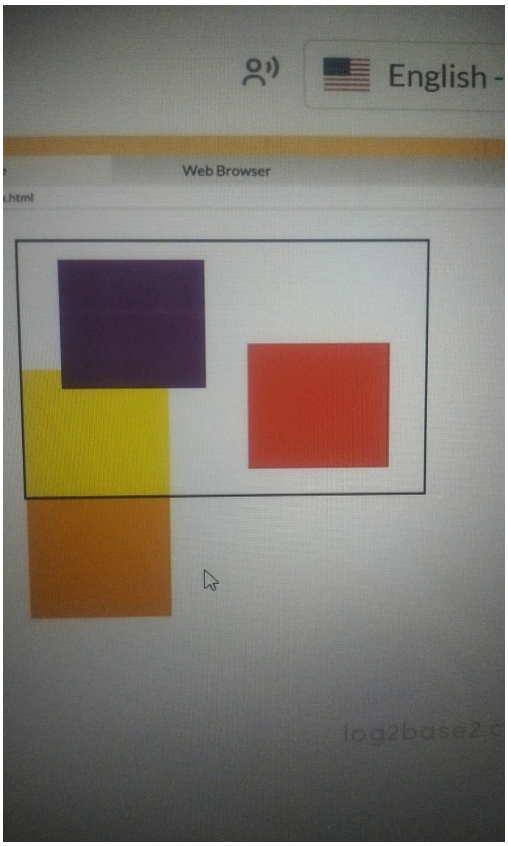
```
    /*  Fixed */
    .orange {
      background-color: orange;
      position: fixed;
      top: 250px;
      right: 50px;
    }

    /*  Absolute  */
    .red {
      background-color: red;
      position: absolute;
      top: 0px;
      right: 50px;
    }

    /*  Sticky   */
    .yellow {
      background-color: yellow;
      position: sticky;
      top: 30px;
    }
  </style>

</head>

<body>
```

```
<div class="container">

  <div class="box purple"></div>
  <div class="box orange"></div>
  <div class="box red"></div>

</div>

<div class="box yellow"></div>

</body>

</html>
```


English
Web Browser
log2base2


English -
My Website
Web Browser
File | index.html

English - US
Web Browser
log2base2.com


English -
Web Browser

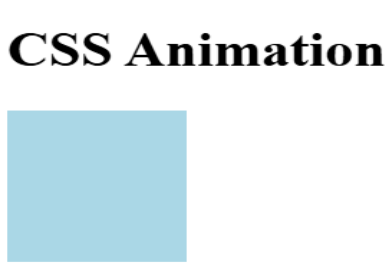
22)Z-Index:

```html
<!DOCTYPE html>
<html>

<head>

  <style>
    div {
      position: absolute;
      width: 175px;
      height: 175px;
      border-radius: 10px;
      color: white;
      padding: 5px;
      text-align: right;
    }

    #first {
      background-color: #FF5C35;
      top: 20px;
      left: 20px;
      z-index: 0;
    }

    #second {
      background-color: #192733;
      top: 50px;
      left: 50px;
      z-index: 1;
```

```
    #third {
      background-color: #4FB06D;
      top: 80px;
      left: 80px;
      z-index: 2;
    }

    #fourth {
      background-color: #ED2D40;
      top: 110px;
      left: 110px;
      z-index: 3;
    }
  </style>

</head>

<body>
  <div id="first">1</div>
  <div id="second">2</div>
  <div id="third">3</div>
  <div id="fourth">4</div>
</body>

</html>
```

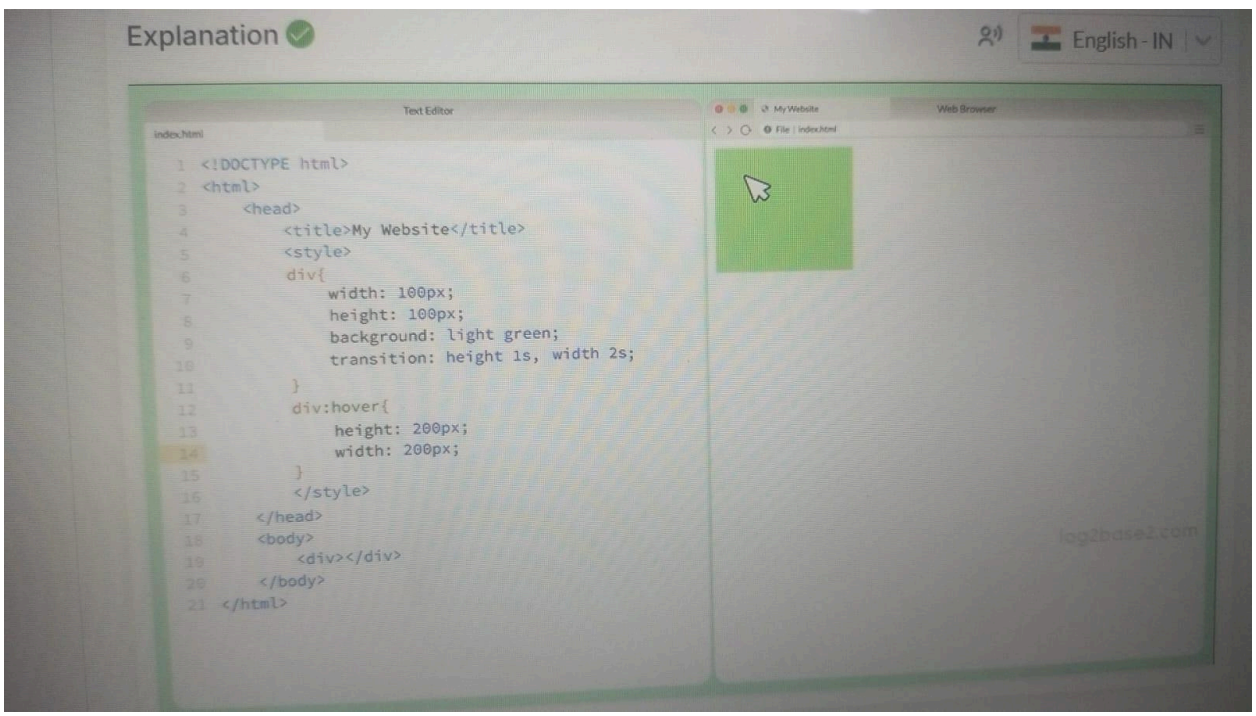
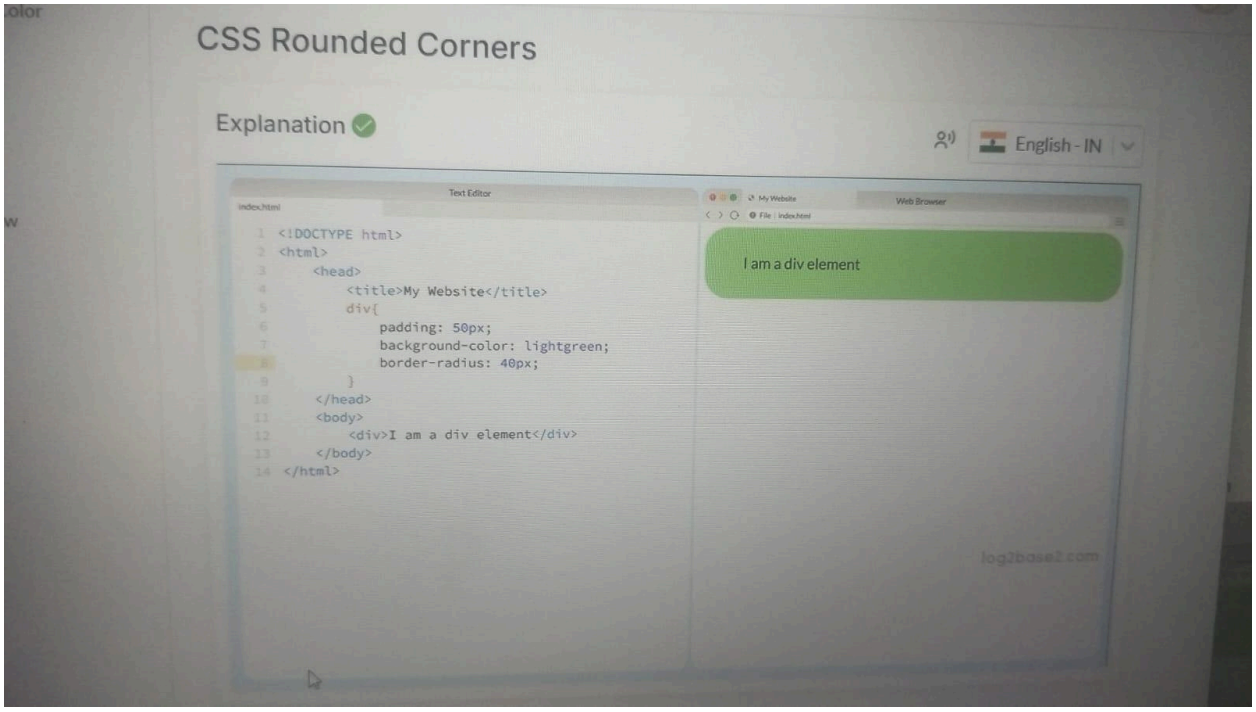
23) css Animations:

```html
<!DOCTYPE html>
<html>
<head>
  <style>
    div {
      width: 100px;
      height: 100px;
      background-color: lightblue;
      animation-name: example1;
      animation-duration: 5s;
    }

    @keyframes example1 {
      0% {  background-color: red;}

      50% { background-color: blue;}

      100% { background-color: green;}
    }
  </style>
</head>

<body>
  <h1>CSS Animation</h1>
  <div></div>
</body>
</html>
```



24) css Transitions: transition: propertyname:duration

If we touch then transition occur



25) css rounded corners: